AYUSH JUVEKAR

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PROFILE

Full-stack developer and ML researcher with an M.S. in Computer Science, production experience as a freelance sole developer, and research depth across computer vision, edge inference, privacy-preserving analytics, and IoT systems.

EDUCATION

Michigan Technological University, Houghton, MI — **CGPA: 3.9/4.0**
M.S. Computer Science — Graduated April 24, 2026

Pune Institute of Computer Technology, Pune, India — **GPA: 8.14/10**
B.E. Computer Engineering — June 2024

EXPERIENCE

The Mind Point - Full-Stack E-commerce Platform | *Next.js 15, React 19, Clerk, Razorpay* — Aug 2025 - Present

- Freelance sole developer for a mental health education course commerce platform with Razorpay payments, Clerk auth, dynamic pricing, BOGO campaigns, persistent carts, and automated enrollment confirmations via Resend
- Built admin tracking with Google Sheets, loyalty points, WCAG accessibility improvements, and PostHog analytics for retention and funnel insights
- Own architecture, implementation, deployment, and release decisions across storefront, checkout, enrollment, admin, analytics, and notification workflows

Teaching Assistant - Physics Department | Michigan Technological University — Sep 2024 - Apr 24, 2026

- Facilitated learning for 60+ students in PH 2100/2200; developed Python automation reducing admin work 20 hours/semester
- Conducted tutorials improving average exam scores 15% and collaborated with faculty on course materials and assessments

SELECTED PROJECTS

YayCamp - Full-Stack Camping Platform | *Next.js, TypeScript, Prisma, PostgreSQL* | yaycamp.ayushjuvekar.com

- Architected full-stack app with Next.js SSR and Clerk authentication managing 100+ users, reducing page load time 40%
- Designed PostgreSQL schema with Prisma ORM, mapped campground discovery with React Leaflet, optimized queries by 30%, and achieved 99.9% uptime

Anomaly Detection in Crowd Surveillance | *Python, TensorFlow, Edge Computing* | CVBDSL Lab

- Trained CNN achieving 92% accuracy on 10K+ frames; optimized with TensorFlow Lite reducing inference time 73% on Jetson Nano
- Deployed real-time pipeline processing 720p at 25 FPS across 5 edge devices, cutting cloud costs 85% and response time 40%

Secure Data Analytics with Homomorphic Encryption | *Python, CKKS, Docker* | CVBDSL Lab

- Built privacy-preserving pipeline using CKKS encryption for healthcare data, achieving 98% accuracy with HIPAA compliance
- Containerized modules with Docker, streamlining deployment and improving processing efficiency 25%

IoT Air Quality Monitoring System | *C++, Arduino, MQTT, Python* | **Published: ESCI 2023**

- Engineered CO monitoring system using MQTT, Zigbee mesh networking, Kalman filtering, and ML predictive alerts to improve detection accuracy 20%
- Presented work at ESCI 2023; publication indexed by IEEE with DOI: 10.1109/ESCI56872.2023.10100144

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C++, Go, Rust, C# | **Frontend:** React, Next.js, TailwindCSS, shadcn/ui | **Backend:** Node.js, Express.js, Prisma ORM, REST APIs | **Databases:** PostgreSQL, MongoDB, Redis | **ML/AI:** TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas | **DevOps:** Docker, Git, Vercel, Linux | **IoT:** Arduino, MQTT, Zigbee | **Auth/Payments:** Clerk, JWT, Razorpay

PUBLICATIONS & AWARDS

A. Juvekar et al., “Carbon Monoxide Concentration Monitoring System,” *ESCI 2023* | **2nd Place** PICT InC 24 (Data Structures) | **Conference Presenter** ESCI 2023 | **Mathex & MSCE Scholarships** 2021-2024